MIL-C-52614B(ME)
13 December 1984
SUPERSEDING
MIL-C-52614A(ME)
7 May 1975

MILITARY SPECIFICATION

CHAIN, BRIDGE ERECTION

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of bridge erection chain.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-C-271	- Chains and Attachments, Welded and Weldless.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-P-116	- Preservation Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3950

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The bridge erection chain shall consist of a grab link, an end link, and a length of close-link chain, and shall be as shown in figure 1 and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3 Physical characteristics.

3.3.1 Chain. The close-link chain shall be in accordance with RR-C-271, type I, grade C, class 4. The nominal link size shall be 5/8 inch.

3.3.2 End link. The end link shall be installed between the close-link chain and the grab link and shall be capable of joining the two. The end link shall have the same physical and general chemical requirements as the close-link chain, except that the strength of the end link may be greater than that of the close-link chain and the nominal end link size shall be 7/8 inch.

3.3.3 Grab link. The grab link shall be in accordance with RR-C-271, type XII, except that the shape and dimensions of the grab link shall be as shown in figure 1.

3.3.4 Length. The overall length of the chain, including the grab link, shall be not less than 14 feet and not more than 14-1/2 feet.

3.4 Identification marking. Identification marking shall be in accordance with MIL-STD-130.

3.5 Workmanship. The close-link chain, end link and grab link shall be free from defects or imperfections that may impair appearance or serviceability.

3.5.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the chain to conform to the examination and inspection requirements specified in section 4.

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3.5.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining tht automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME and ANSI/AWS codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. All the bridge erection chains offered for delivery at one time shall constitute a lot.

4.3.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined as specified in 4.4.1. Acceptable quality level (AQL) shall be 1.0 percent defective for major defects and 2.5 percent defective for the minor defect.

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4.4 Inspection procedure.

4.4.1 Examination. The bridge erection chains shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Chain link not as specified.
103. End link not as specified.
104. Grab link not as specified.
105. Close-link chain length not as specified in figure 1.
106. Workmanship not as specified.
107. Welding not as specified.
108. Welders certification not made available for review.

Minor

201. Identification marking not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level S-2.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109. Materials not as specified for preservation, packaging and packing.
110. Surfaces not cleaned as specified for level A.
111. Application of preservaties not as specified for level A.
112. Securing wire not as specified for level A.
113. Packing not as specified.
114. Marking illegible, incomplete or incorrect.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The surface of each chain shall be cleaned in accordance with MIL-P-116, method C-1, until free from rust, scale and other foreign matter, and coated with type P-1 preservative. Each length of chain shall be coiled and each end of the chain secured with soft annealed corrosion-resistance wire, twisted and tucked under to avoid snagging.

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5.1.2 Level C. The chains shall be preserved and packaged in a manner to afford protection against damage from the contractor to the initial destination. The contractor's standard practice shall be acceptable provided it fulfills these requirements.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The chains shall be packed in close-fitting cleated-plywood boxes conforming to PPP-B-601, overseas type, style optional, up to the weight limitation of the container. The boxes shall be closed and strapped in accordance with the appendix to the box specification. On modified boxes, strapping shall be placed adjacent to the skids.

5.2.2 Level B. The chains shall be packed as specified in 5.2.1, except that the boxes shall be domestic type, style optional.

5.2.3 Level C. The chains shall be packed in containers to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bridge erection chain is intended for use with bridge erection equipment for single-lane fixed panel bridges.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

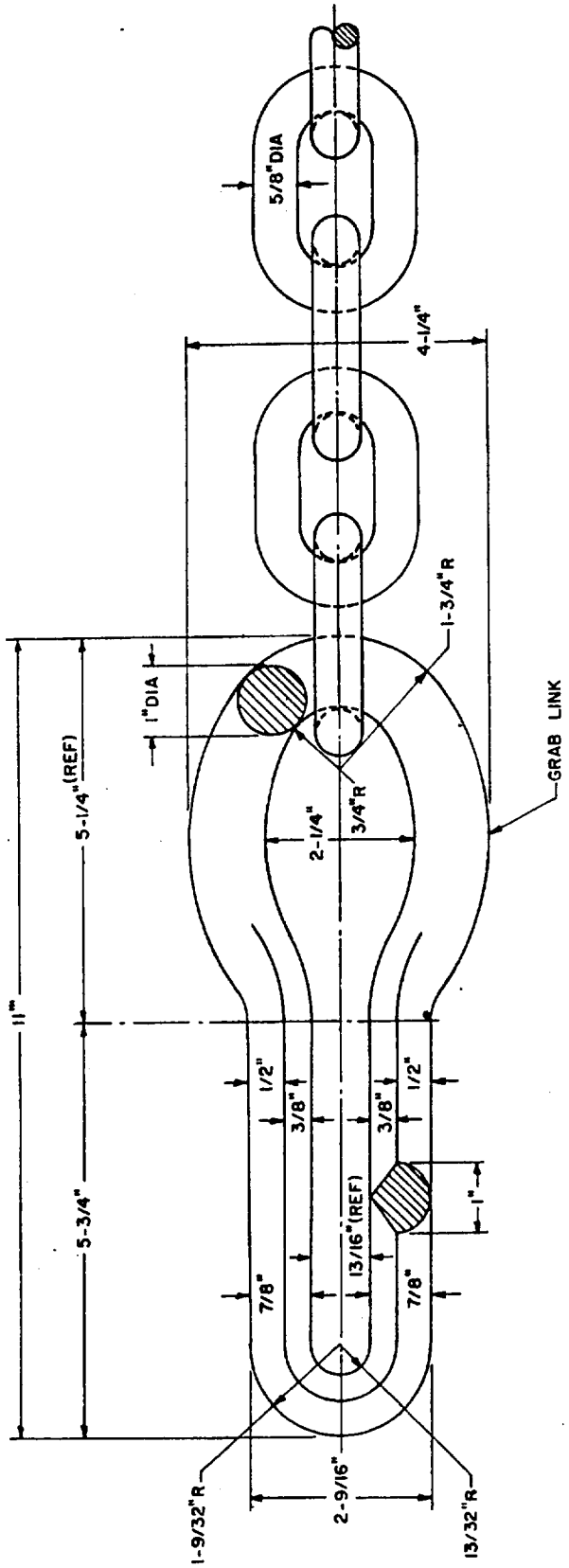
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FIGURE 1. Chain assembly: bridge erection.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52614B(ME)	Chain, Bridge Erection

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
3b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
7c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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